

EXTRA CHALLENGE QUESTIONS (C REVISION PHASE SUPPLEMENT - 10 ITEM EXTENSION)

No.	Course Content	Planning Title	Planning Description
2E	Revision of C	Demonstrate the use of loops and functions from the following C Programs.	8a. WAP to print the multiplication table of a given number N using a loop. (A) e.g., Input N = 3 -> Output: 3x1=3, 3x2=6... up to 3x10=30
			8b. WAP to find the sum of all odd numbers from 1 to N using a loop. (A) e.g., Input N = 5 -> Output: Sum = 9 (1 + 3 + 5)
			8c. WAP to count the total number of digits in a given integer. (A) e.g., Input: 5241 -> Output: Total digits = 4
			8d. WAP to print the following simple number pattern using nested loops. (B) 1 22 333 4444
			8e. WAP to find the sum of all individual digits of a given number. (A) e.g., Input: 123 -> Output: Sum = 6 (1 + 2 + 3)
			8f. WAP to calculate the power of a number (X^Y) using a basic while/for loop. (A) e.g., Input: X=2, Y=3 -> Output: 8
			8g. WAP to print all even numbers between 1 and user-defined number N. (A) e.g., Input N = 6 -> Output: 2, 4, 6
			8h. WAP to check whether a given year is a leap year or not using a custom function. (B) e.g., Input: 2024 -> Output: Leap Year
			8i. WAP to print the reverse of a given integer using a loop. (B) e.g., Input: 456 -> Output: 654
			8j. WAP to find the sum of first N natural numbers using a standard loop. (A) e.g., Input N = 4 -> Output: 10 (1 + 2 + 3 + 4)
3E	Revision of C	Demonstrate the use of arrays using following C Programs.	12a. WAP to calculate the sum and average of all numbers stored inside a 1D array. (A) e.g., Input: {2, 4, 6} -> Output: Sum = 12, Average = 4
			12b. WAP to copy all values of one 1D array into a second empty array item-by-item. (A) e.g., Input: Source={5, 9} -> Output: Destination={5, 9}
			12c. WAP to add two 1D arrays of equal sizes element-by-element and store in a third array. (B) e.g., {1, 2} + {10, 20} -> Output: {11, 22}
			12d. WAP to calculate the combined sum of all elements in a standard 3 X 3 matrix. (B) e.g., Computes total value of all 9 elements inside the grid layout.
			12e. WAP to find the maximum element from a 1D array of size N. (A) e.g., Input: {4, 9, 2} -> Output: Max = 9
			12f. WAP to count how many times a user-specified number X appears in a 1D array. (A) e.g., Input: {1, 3, 3, 5}, X=3 -> Output: Occurs 2 times
			12g. WAP to print all elements of a 1D array in exact reverse order. (A) e.g., Input: {7, 8, 9} -> Output: 9, 8, 7
			12h. WAP to display only the diagonal elements of a 3 X 3 matrix layout. (A) e.g., Prints positions at indexes [0][0], [1][1], and [2][2] directly.
			12i. WAP to check if two separate 1D arrays are identical (all indexes contain matching elements). (B) e.g., Checks if array size and elements align exactly index-for-index.
			12j. WAP to count the total count of even numbers inside a 3 X 3 matrix. (B) e.g., Scans grid and returns quantity of values cleanly divisible by 2.
			16a. WAP to find the total length of a string without using the built-in strlen() function. (A)

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4E	Revision of C	Implement the following programs using String Functions in C.	e.g., Input: "book" -> Output: Length = 4 16b. WAP to count the total number of blank spaces present in a given string line. (A) e.g., Input: "my code template" -> Output: Total spaces = 2 16c. WAP to copy one string into another character-by-character without using strcpy(). (B) e.g., Input: "data" -> Output: Target String copy = "data" 16d. WAP to find the first index position of a specific character inside a string. (B) e.g., Input String: "ball", Find: 'l' -> Output: First Index found at 2 16e. WAP to count the total number of vowels (a, e, i, o, u) inside a given string. (A) e.g., Input: "open" -> Output: Vowels = 2 16f. WAP to concatenate two strings manually without using the built-in strcat() function. (B) e.g., Input: "sub" + "set" -> Output: "subset" 16g. WAP to verify if two strings match without using the built-in strcmp() utility. (B) e.g., Input: "yes" and "yes" -> Output: Strings are Equal 16h. WAP to copy a string into another character array in reverse order manually. (B) e.g., Input: "abc" -> Output: "cba" 16i. WAP to count total alphabets and total numeric digits separately in an alphanumeric string. (B) e.g., Input: "code12" -> Output: Alphabets = 4, Digits = 2 16j. WAP to replace every empty blank space character in a string with a hyphen ('-'). (B) e.g., Input: "a b c" -> Output: "a-b-c"